What does a closed Suez Canal do to our chair line, and does a German supplier fix it?

Baseline, Suez closed, Suez closed with a German supplier: the line stands still **26.8 → 133.1 → 8.3 hours** a year, makes **699.8 → 690.6 → 699.8 chairs** a week and emits **4.60 → 4.63 → 4.19 kg CO_2e** per chair. The closure costs about 480 chairs a year; the German supplier from week 16 removes that loss but makes a chair 2.6 € dearer.

Baseline
Suez closed (Red Sea diversion)
Suez closed + German supplier

1 Product and line

The Interstuhl EVERY POLSTER Aktionsmodell EV15R: five stations and one key part.

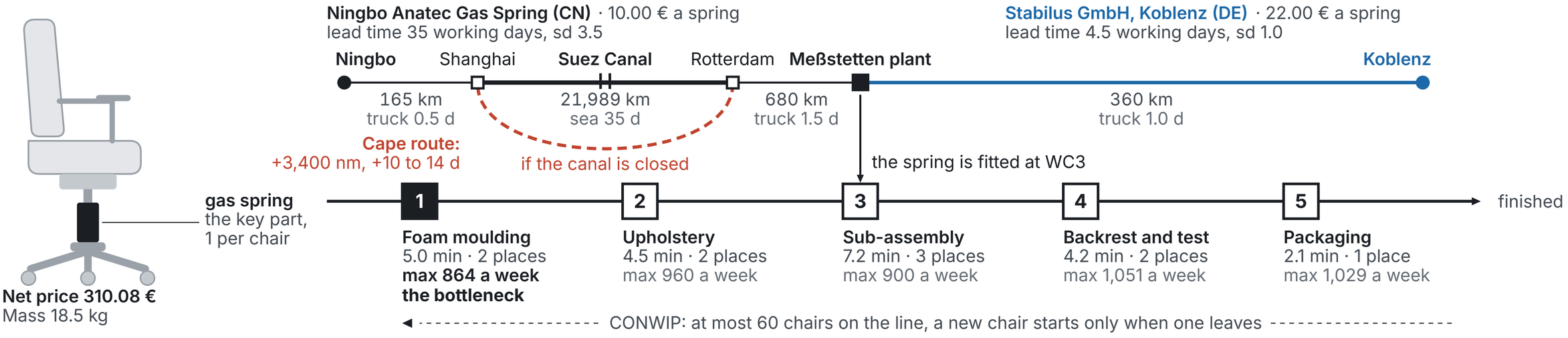


Fig. 1 routing.csv, workcenters.csv (capacity includes availability, 90 %, WC4 92 %), network_lanes.csv (d = calendar days), suppliers.csv (lead times in working days: Ningbo 48 calendar days × 5/7 = 34.3, set to 35).

2 The disruption and the event behind it

Houthi attacks on merchant ships began in November 2023. Carriers left the Red Sea and the Suez Canal and sailed round the Cape of Good Hope instead.

15 Dec 2023	Maersk and Hapag-Lloyd halt their Red Sea passages[1]
2 Jan 2024	Maersk pauses all Red Sea transits until further notice[2]
29 Jan – 11 Feb 2024	Tesla stops most output in Grünheide for lack of components[3]
Each voyage	The Cape route adds 10–14 days[4] and 3,400–3,600 nautical miles[5]

IN THE MODEL

Window	Weeks 15–26 (12 weeks), our assumption. A shipment whose Suez passage (65 % into its lead time) falls in it goes round the Cape.
Response	From week 16 new orders go to Stabilus; shipments on their way still arrive.
Demand	700 chairs a week; backlog of at most 700, beyond that demand is lost.
Stood still	A chair waits at WC3, no spring in stock, no WC3 workplace busy. Merely waiting demand does not count.
Not used	Ever Given 2021: six days, absorbed by the safety stock.

SOURCES [1] Lloyd's List (2023) [2] Automotive Logistics (2024) [3] Reuters (2024) [4] Trans.info (2023), Automotive Logistics (2024) [5] C.H. Robinson (2024), gCaptain (2024). Full references in the notebook.

3 Scoreboard

	Chairs per week	Total cost per chair, €	CO_2e per chair, kg	Line stood still, hours a year
Baseline	**699.8** 699.8 – 699.9	**149.93** 149.93 – 149.94	**4.60** 4.60 – 4.60	**26.8** 0.0 – 53.7
Suez closed	**690.6** −9.2 679.0 – 699.8	**150.49** +0.56 150.24 – 150.89	**4.63** +0.03 4.62 – 4.64	**133.1** +106.3 98.6 – 217.6
+ German supplier	**699.8** ±0.0 699.8 – 699.9	**152.55** +2.61 152.54 – 152.55	**4.19** −0.41 4.18 – 4.19	**8.3** −18.5 0.0 – 42.9

Mean of 10 seeds (same random numbers in all scenarios); below it the lowest – highest seed, beside it the change against the baseline. Cost = fixed + material + freight + holding + CO_2e at 100 € a tonne; it depends on supplier prices and freight rates (panel E). Electricity counts in CO_2e but not in euros (no price in the data); it is nearly the same in all scenarios.



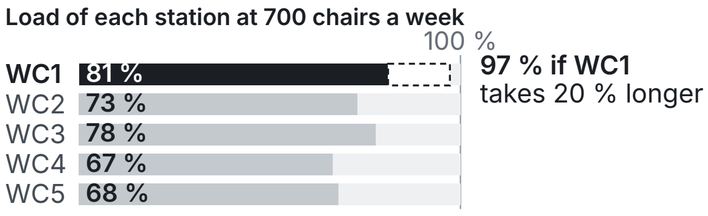
Fig. 2 Chairs finished per week. With Stabilus the line makes 700 a week again from week 17.

4 Six course ideas that explain the numbers

A Bottleneck

Why does catching up take so long?

Load of each station at 700 chairs a week

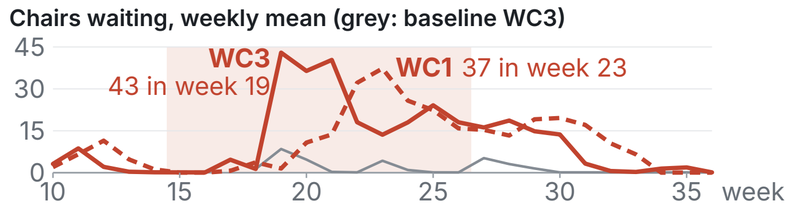


The slowest station limits the whole line. WC1 finishes at most 864 chairs a week, so against 700 demanded only 164 a week are spare. After the restart the output peaks at 828 (week 22) and cannot rise much further.

B Variability and waiting

Why do chairs queue although no station is at 100 %?

Chairs waiting, weekly mean (grey: baseline WC3)

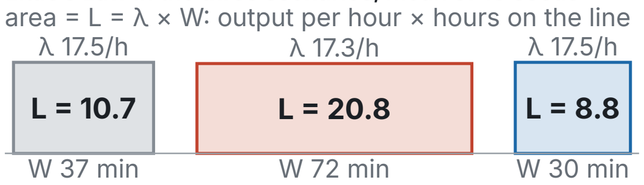


Variability creates queues even below 100 % load. Without springs, chairs pile up at WC3 (43 in week 19). When the springs return, the burst reaches WC1, which has only 164 chairs a week to spare: its queue peaks at 37 in week 23.

C Little's Law

Why does time on the line double while output barely falls?

Little's Law in the three scenarios, mean of 10 seeds
area = L = λ × W: output per hour × hours on the line

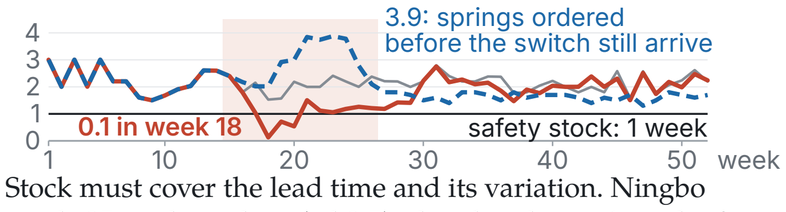


L = λ × W: chairs on the line = output per hour × hours per chair. Output hardly changes (17.3 chairs an hour against 17.5), but a chair needs 72 minutes instead of 37, so 20.8 chairs are on the line instead of 10.7.

D Lead time and safety stock

Why does a detour of 10 to 14 days stop the line?

Weeks of spring cover in stock (stock ÷ weekly demand)
Baseline — Suez closed — German supplier

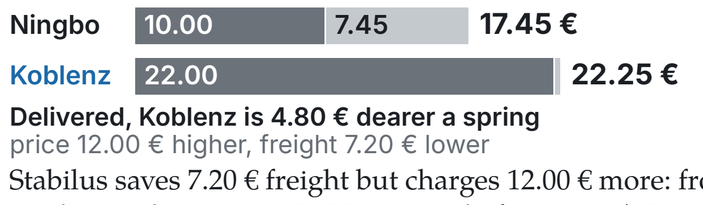


Stock must cover the lead time and its variation. Ningbo needs 35 working days (sd 3.5), the plant keeps 1 week of safety stock, and the Cape adds 7 to 10 working days: the springs ran out in all ten seeds. The baseline stops too (26.8 h); without the lead-time variation it would not.

E Total cost against unit price

Why does the German supplier make a chair dearer?

Delivered cost of one gas spring, €



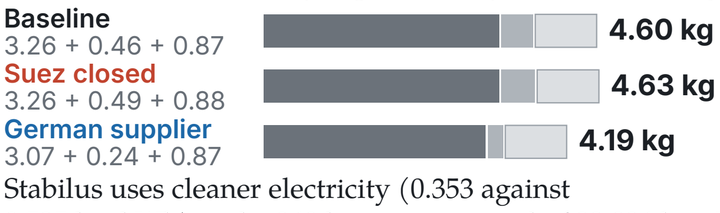
Delivered, Koblenz is 4.80 € dearer a spring
price 12.00 € higher, freight 7.20 € lower

Stabilus saves 7.20 € freight but charges 12.00 € more: from week 16 a chair costs 152.55 € instead of 149.93 € (150.49 € without switch). Ningbo stays normal because it is cheaper; Stabilus buys output. Break-even 17.2 €; 22.00 € is our estimate. Not modelled: contracts, quality, capacity.

F CO_2e per unit

Why does CO_2e per chair fall by 9 %?

CO_2e per chair, kg



Stabilus uses cleaner electricity (0.353 against 0.525 kg/kWh) and a 360 km route instead of 22,834 km. It supplies 54 % of the year's springs, so per chair 0.19 kg are saved on the spring and 0.22 kg on transport.

5 Two papers behind the interpretation

Ambulkar, Blackhurst & Grawe (2015)

QUESTION How does a firm become resilient to supply chain disruptions? The authors build a scale for firm resilience and test what drives it.

FINDING Disruption orientation alone is not enough. After a high-impact disruption its effect on resilience runs entirely through reconfiguring resources; after a low-impact one it works together with risk-management infrastructure.

EXPLAINS the fall of the stand-still from 133 to 8 hours. We treat the closure as high-impact: the spring stock runs out in all ten seeds. The switch to Stabilus in week 16 is the reconfiguration; from then on the line stands still for 0.1 hours instead of 125.

Schoenherr, Mena, Vakil & Choi (2023)

QUESTION How can measuring risk make supply chains more resilient? The authors interviewed purchasing and supply professionals and propose a framework for measuring.

FINDING Risk must be measured continuously and the numbers must lead to action: a culture of measuring, not an occasional check.

EXPLAINS why the switch in week 16 is in time. Weeks of spring cover is such a measure: 2.4 in week 15, 1.8 in week 16, 0.1 in week 18. A plant that tracks it acts in week 16; Stabilus delivers in 4.5 days, inside the 1.8 weeks left. Stand-still hours peak in weeks 19 to 21.

REFERENCES

Ambulkar, S., Blackhurst, J., & Grawe, S. (2015). Firm's resilience to supply chain disruptions: Scale development and empirical examination. *Journal of Operations Management, 33–34*, 111–122. https://doi.org/10.1016/j.jom.2014.11.002

Schoenherr, T., Mena, C., Vakil, B., & Choi, T. Y. (2023). Creating resilient supply chains through a culture of measuring. *Journal of Purchasing and Supply Management, 29*(4), 100824. https://doi.org/10.1016/j.pursup.2023.100824